Luiz Cauet Castilla Ruy Blum

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EDUCATION

University of California, Santa Cruz — Santa Cruz, California
B.S. in Computer Engineering - Systems Programming, CS Minor — *Expected Graduation, March 2027*

GPA: 3.5

Related Coursework: Machine Learning, Data Structures & Algorithms, Networks, Systems Design, Computer Architecture, Software Engineering, Organization, Game AI, Mechatronics

EXPERIENCE

Augmented Design Lab — Santa Cruz, California
Research Participant (Advisor: Dr. Jim Whitehead) — Jan 2026 – Mar 2027

- Developed PCG prototypes using LLMs, Wave Function Collapse, and Z3 constraint solving
- Implemented grid-based world generation systems and constraint-driven content pipelines
- Analyzed and applied technical research in procedural generation and game systems

The Audition — Santa Cruz, California
Lead AI & Software Engineer — September 2025 – Current

- Leading a 13-member multidisciplinary team while serving as the project's lead software engineer.
- Designed, trained, and evaluated a PyTorch model to classify guitar solo quality, achieving 85% agreement with human rankings using Spearman rank correlation.
- Architecting and implementing gameplay systems, AI behaviors, and core game infrastructure in Godot.

Integem Inc. — Fremont, California
AI & Robotics Instructor — June 2026 – July 2026

- Taught robotics, computer vision, and generative AI concepts to students ages 10–17
- Guided students through the complete YOLOv5 training pipeline, including dataset creation, annotation, training, evaluation, and hyperparameter tuning.
- Instructed deploying of pretrained vision models for robotic object detection and manipulation workflows.
- Mentored students through NVIDIA's Generative AI curriculum, introducing diffusion models and practical machine learning concepts.

The Slug Collective — Santa Cruz, California
Founder & President — Dec 2024 – Current

- Founded and led a 100+ member interdisciplinary music and tech collective at UCSC
- Secured $1,500 in funding through technical project proposals and organizational leadership
- Directed collaborative production of a 12-track multi-contributor digital release

PROJECTS

Resonance - Text-to-Synth Preset Generator (ML, LLM, Python, Next.js) — 2026

- Developed a multimodal ML pipeline mapping natural language prompts to synthesizer presets using CLAP audio-text embeddings and vector similarity search
- Experimented with LLM-driven iterative fine-tuning of preset parameters based on user feedback
- Implemented Python + FastAPI backend services for embedding generation, retrieval, and preset manipulation; containerized ML pipelines on Oracle Cloud VM with Docker.
- Built a Next.js interface enabling real-time semantic search and interactive preset modification

SmartQueue (Next.js, React, TypeScript, Supabase, Spotify SDK) — 2025

- Built an adaptive queue system that dynamically reorders playlists based on user behavior
- Designed a weighted recommendation algorithm with persistent canonical state using Supabase Auth and RLS

Music Collaboration Forum (Node.js/Express.js, AWS S3, React) — 2024

- 1st place hackathon winner. Built a full-stack collaboration platform supporting authentication, media uploads, and real-time user interaction
- Implemented AWS S3-backed audio storage and scalable backend services

SKILLS

Languages: Python, C/C++, TypeScript, SQL, Java, Assembly
AI/ML: PyTorch, CUDA, NumPy, SciPy, FastAPI
Infrastructure: Linux, Docker, PostgreSQL, AWS, Supabase, Wireshark